Imbalance Mitigation for TCGA-UT Across Patch and WSI-Bag Regimes

Abstract

Class imbalance in computational pathology depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. This study evaluates imbalance mitigation on TCGA-UT with two separate native-regime benchmarks on shared slide-disjoint splits and three seeds. On the patch benchmark with frozen Virchow2 embeddings, Scholz-style metric-aligned losses provide small but consistent gains over plain cross-entropy, whereas ProGAN augmentation and focal loss do not. On the WSI-bag benchmark, RankMix shows a clearer benefit, especially in balanced accuracy, with more modest macro-F1 gains and overlapping seed variability. Imbalance mechanisms are therefore regime-dependent rather than universally transferable across patch and slide-level tasks.

1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive summaries. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware summaries as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

A common source of ambiguity in pathology imbalance studies is the input regime. Some methods are naturally image-level methods: they assume labeled images, image embeddings, or directly generated image samples. Other methods are naturally WSI-level methods: they assume weak labels over bags of instances and use attention, instance ranking, or bag-level representation learning. Comparing such methods in one pooled leaderboard can confound the imbalance mechanism with changes in supervision, input size, backbone, and aggregation.

TCGA-UT is useful for separating these questions because it supports two legitimate tasks with different native inputs. The dataset provides labeled histopathology patches cropped from pathologist-selected tumor regions and slide identities that can be grouped into WSI-level bags. Patch labels are class labels, not dense pixel masks or cell-level annotations. This study therefore uses the available supervision directly: patch-level methods are evaluated on controlled patch classification with frozen Virchow2 embeddings, while MIL methods are evaluated on slide-level feature bags built from the same foundation-model family.

The contribution is a controlled benchmark design rather than a reproduction claim for every cited method. The study uses one dataset, two native tasks, and one controlled comparison within each task. Method fidelity is defined by implementing the defining imbalance mechanism in the regime for which that mechanism was designed, while holding the backbone family, split, and evaluation code fixed within each benchmark.

2 Dataset

TCGA-UT is derived from diagnostic *whole-slide images* (WSIs) in The Cancer Genome Atlas (TCGA), a pan-cancer resource that links molecular and clinical cancer data with large image collections (Tomczak et al., 2015). The TCGA-UT release turns this slide archive into a computational-pathology image dataset by extracting hematoxylin and eosin (H&E) tumor *patches* from pathologist-selected *uniform tumor regions*. It contains 1,608,060 RGB patches of size 256×256 pixels from 8736 diagnostic slides and 32 *cancer-type labels*. The patch labels are slide- or cancer-type labels, not dense tissue masks or cell-level annotations (Komura et al., 2022).

This structure makes TCGA-UT well suited for an imbalance study because the imbalance is native to the dataset rather than imposed only after sampling. The largest class is glioblastoma multiforme (GBM; 792 slides), while the smallest class is diffuse large B-cell lymphoma (DLBC; 28 slides). The resulting slide-level imbalance ratio is 28.3. As a consequence, the dataset exposes the practical tension that motivates this study: methods must learn from many visually related cancer categories while preserving performance on classes with far fewer slide-level examples.

Table 1: Dataset summary used by the shared patch and WSI-bag benchmarks.

Property	Value
Cancer-type labels	32
Diagnostic slides	8736
TCGA-UT patches	1,608,060 patches of 256×256 pixels
Largest class	Glioblastoma multiforme (GBM), 792 slides
Smallest class	Diffuse large B-cell lymphoma (DLBC), 28 slides
Slide-level imbalance ratio	28.3
Shared split size per seed	6116 train, 1310 validation, 1310 test slides
Controlled patch input	Resolution level 0; up to 30 deterministic patches per slide; one frozen Virchow2 embedding per patch (Section 2.2)
WSI-bag input	Uncapped frozen Virchow2 feature bags from the same slide identities

2.1 Data Units and Terminology

Several data units are involved in the benchmark, and the distinction matters for method comparison. A *slide* or *WSI* is one diagnostic histopathology image associated with one cancer-type label. A *tumor region* is a pathologist-selected area of a slide from which TCGA-UT crops groups of patches. A *patch* is one 256×256 RGB crop selected by the controlled manifest. A *patch embedding* is the frozen Virchow2 vector computed from that crop and consumed by the patch-level linear classifier. A *feature chunk* is a saved frozen-feature representation associated with a slide or slide subset. A *WSI bag* is the collection of frozen patch-feature instances used by the attention-MIL model for one slide-level prediction. *Head*, *body*, and *tail* classes refer to fixed support tiers derived from the full TCGA-UT slide distribution when reporting aggregate class-group metrics on the held-out test split. A *slide-disjoint split* means that all patches and features from one slide stay in exactly one of train, validation, or test.

2.2 Shared Splits and Benchmark Inputs

For each seed, slides are split into 6116 training, 1310 validation, and 1310 test slides. The same slide-level assignments are reused by both benchmarks so that no patches from the same slide can appear in different splits. This is the main leakage-control mechanism: the patch benchmark and the WSI-bag benchmark see different input units, but they inherit the same train/validation/test slide partition.

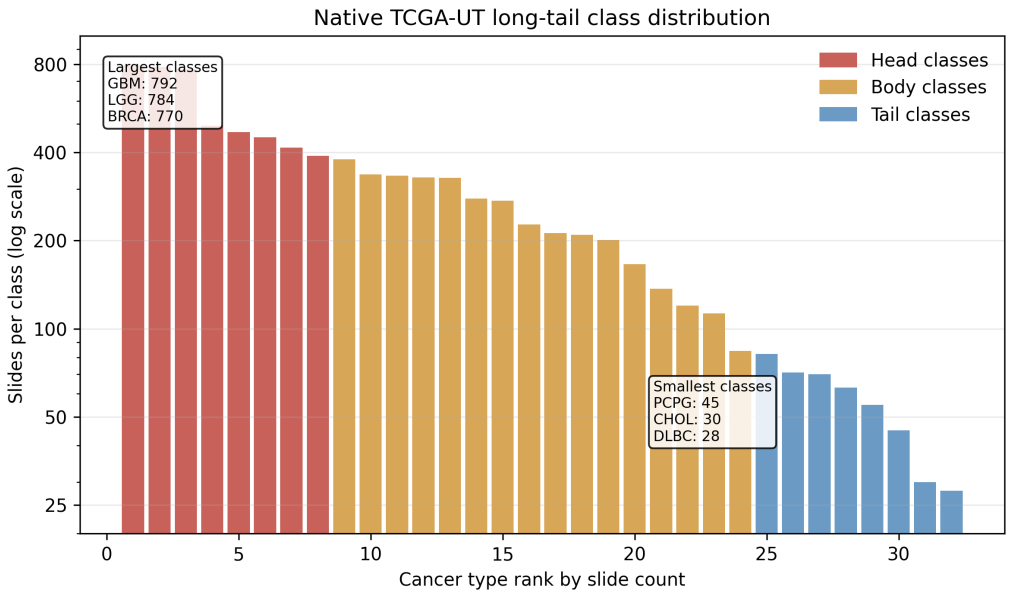
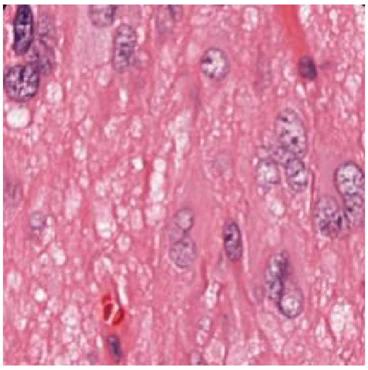


Figure 1: Native TCGA-UT slide support by cancer-type rank. The long-tailed distribution motivates metrics that weight classes more evenly than aggregate accuracy.



Field	Example value
Cancer type	Glioblastoma multiforme
Resolution level	0
Patch size	256×256 RGB pixels
Source unit	One crop from a selected tumor region
Patch role	Image-level sample if selected by the controlled manifest
WSI-bag role	Candidate instance in the slide's feature bag

Figure 2: Example TCGA-UT data point copied from the shared raw dataset. The image is a single patch, while the table shows how that patch relates to the manifest fields used by the benchmark.

The patch benchmark derives a deterministic controlled patch manifest from the shared slide split. It uses resolution level 0 and samples up to 30 patches per slide with a fixed rule before method-specific training begins. The cap is a compromise between data coverage and slide-level control: it uses the empirical median number of raw JPG patches per slide at resolution level 0 (median 30, mean 31.1, range 10–80) while preventing slides with many extracted tumor-region patches from dominating the patch-level loss. At this resolution, 8686 of 8736 slides have at least 30 raw patches, so the cap retains the typical per-slide patch evidence while limiting only the larger slide folders. Each selected patch is encoded once with a frozen Virchow2 model ('hf-hub:paige-ai/Virchow2' via `timm`), and all patch methods train the same lightweight classifier head on these embeddings (Zimmermann et al., 2024; Vorontsov et al., 2024). The encoder output is concatenated as $\mathrm{concat}(\mathrm{CLS}, \mathrm{mean}(\text{patch tokens}))$, yielding 2560-dimensional float16 vectors that are cast to float32 before classifier training. This design isolates imbalance mechanisms from both patch-selection policy and end-to-end image-backbone learning. The WSI-bag benchmark uses full frozen Virchow2 feature bags built from the same slide identities, rather than applying an additional instance cap.

Table 2: Terminology used for the two native input regimes.

Term	Meaning in this study
Slide / WSI	Diagnostic whole-slide image carrying one cancer-type label.
Patch	256×256 H&E crop from a selected tumor region; patch labels inherit the slide cancer type.
Patch embedding	Frozen Virchow2 vector for one controlled patch; input to the patch-level classifier head.
Region	Tumor area selected before patch extraction.
Cancer type	The 32-way class label used for patch and slide-level classification.
Feature chunk	Frozen Virchow2 feature representation used to assemble slide-level feature bags.
WSI bag	Set of patch-feature instances for one slide, consumed by the MIL model.
Head/body/tail class	Fixed support tier from the full TCGA-UT slide distribution; see Section 3.8.
Slide-disjoint split	Split rule preventing any slide's patches or features from crossing train, validation, and test.

3 Methods

The methods are organized by the level at which they intervene in the imbalanced learning problem. Cross-entropy (CE) is treated as the no-mitigation reference condition: it uses the observed training distribution as given and does not rebalance samples, labels, costs, representations, or generated data. The remaining methods are tested imbalance interventions grouped according to the taxonomy used in the class-imbalance literature: data-level sampling and augmentation, algorithm-level losses, hybrid sampling–loss methods, synthetic generation, and representation or architecture-level methods (Saini and Susan, 2023; Zhang et al., 2023).

The comparison is regime-specific. Patch methods consume frozen Virchow2 embeddings of controlled patches and optimize one shared linear classifier head. WSI-bag methods consume full frozen Virchow2 feature bags and optimize one shared attention-MIL head. Both benchmarks therefore rely on the same pathology foundation-model family, but they differ in supervision unit: one embedding per selected patch versus aggregation over all instances on a slide. Method differences within a benchmark therefore correspond to imbalance mechanisms rather than changes in split, evaluation code, or encoder choice.

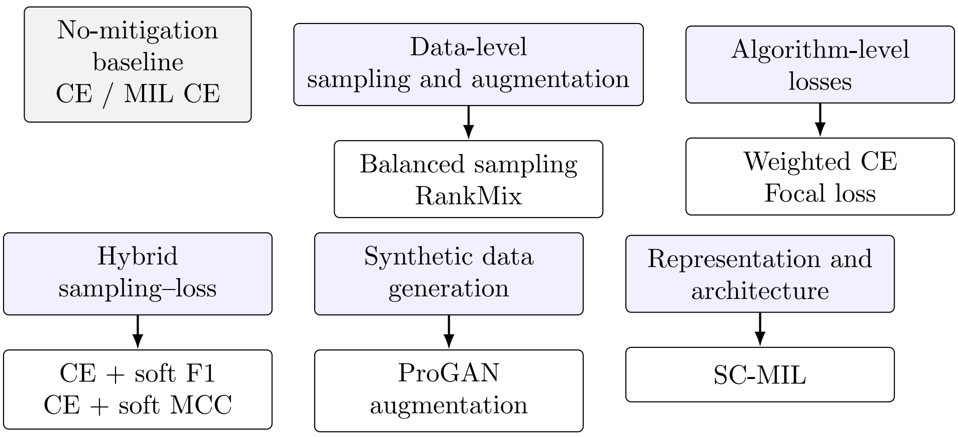


Figure 3: Method organization used in the benchmark. Cross-entropy is the no-mitigation reference, while the tested alternatives intervene through sampling, loss design, synthetic data, WSI feature augmentation, or representation learning.

3.1 Notation

Let C be the number of cancer-type classes and let n_c denote the number of training examples from class c. For a labeled patch or slide-level bag with label $y \in \{1, \dots, C\}$, the model outputs logits $\mathbf{z} \in \mathbb{R}^C$ and class probabilities

$$p_c = \frac{\exp(z_c)}{\sum_{j=1}^{C} \exp(z_j)}. \tag{1}$$

For WSI-bag methods, a slide is represented as a bag $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ of frozen patch-feature instances. The attention-MIL backbone maps each bag to a bag embedding $\mathbf{b}(B)$ and class logits. Unless a method explicitly changes the sampler, objective, or representation loss, the empirical class frequencies in the training set define the optimization signal.

3.2 Cross-Entropy Baseline

Unweighted cross-entropy is the baseline because it is the standard supervised objective without an explicit class-imbalance correction. For an example with label y, the loss is

$$\mathcal{L}_{\mathrm{CE}}(\mathbf{z}, y) = -\log p_y. \tag{2}$$

In the patch benchmark, this objective trains the shared linear head on frozen Virchow2 patch embeddings from the controlled manifest. In the WSI-bag benchmark, the same objective trains the shared attention-MIL model on slide-level feature bags. In both cases, CE uses the observed class distribution directly: majority classes appear more often in minibatches and therefore contribute more loss terms over an epoch. This makes CE an appropriate reference for asking whether an imbalance-specific intervention actually improves the rare-class tradeoff.

3.3 Data-Level Sampling and Augmentation

Data-level methods change what the optimizer sees, rather than changing the mathematical penalty assigned to one observed example. In the taxonomy, this family includes resampling, augmentation, and WSI-specific feature mixing strategies that make minority or informative examples more visible during training (Saini and Susan, 2023).

3.3.1 Balanced Sampling

Balanced sampling changes minibatch composition by sampling training examples with probabilities inversely related to their class support. For an example i with class y_i, the sampler weight is proportional to

$$q_i \propto \frac{1}{n_{y_i}}. \tag{3}$$

After normalization across the training set, this makes the expected class exposure closer to uniform even though the underlying dataset remains long-tailed. The patch variant applies this sampler to patch examples and still optimizes CE. The MIL variant applies the same idea at slide-bag level and also optimizes CE. Balanced sampling therefore tests a pure data-level intervention: it asks whether equalizing training exposure is enough without changing the loss formula.

3.3.2 RankMix

RankMix is a WSI-native data-augmentation method for weakly supervised slide classification. Ordinary image mixup assumes two inputs with compatible spatial shapes, but WSI bags can contain different numbers of feature instances and lack instance-level labels. RankMix addresses this by mixing representative feature subsequences rather than raw slides (Chen and Lu, 2023).

The tested implementation follows the defining two-stage idea with $\alpha = 1.0$ in $\mathrm{Beta}(\alpha, \alpha)$. First, a teacher attention-MIL model is trained with slide labels for 30 epochs using the same MIL architecture and optimization settings as plain MIL CE; when available, the matched plain MIL CE checkpoint for the same seed is reused as the teacher. For each bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ and target class, the teacher assigns ranking scores to instances using its softmax probability for that class; the top-k instances are kept as a representative subsequence, where $k = \min(|B_a|, |B_b|)$ for the two bags being mixed, and their original within-bag order is preserved after selection. For two bags a and b, the selected feature matrices H_a and H_b are mixed with $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$:

$$\widetilde{H} = \lambda H_a + (1 - \lambda) H_b, \qquad \widetilde{\mathbf{y}} = \lambda \mathbf{y}_a + (1 - \lambda) \mathbf{y}_b. \tag{4}$$

The student MIL model is then trained with soft-label cross-entropy on the mixed bags. This mechanism is categorized as data-level augmentation because it increases the diversity of the training bags in feature space while retaining the same attention-MIL prediction architecture.

3.4 Algorithm-Level Losses

Algorithm-level methods keep the training examples fixed but alter how errors are penalized. They are cost-sensitive or difficulty-sensitive objectives: minority examples may receive larger explicit weights, or easy examples may contribute less to the gradient (Saini and Susan, 2023; Scholz et al., 2024).

3.4.1 Class-Weighted Cross-Entropy

Class-weighted CE multiplies each example's CE term by a class-dependent cost:

$$\mathcal{L}_{\mathrm{WCE}}(\mathbf{z}, y) = -w_y \log p_y, \qquad w_c \propto \frac{1}{n_c}. \tag{5}$$

The implementation normalizes the inverse-frequency weights so that the average scale of the loss remains comparable across methods. This method leaves the sample order and model architecture unchanged, but increases the loss contribution of classes with fewer training examples. It is evaluated for both patch classification and attention-MIL.

3.4.2 Focal Loss

Focal loss reduces the contribution of confidently classified examples and concentrates optimization on examples that remain difficult. With focusing parameter γ, the multiclass focal objective is

$$\mathcal{L}_{\mathrm{focal}}(\mathbf{z}, y) = -(1 - p_y)^{\gamma} \log p_y. \tag{6}$$

This study uses $\gamma = 2.0$ for the patch and WSI-bag variants. The method is relevant to imbalance because many minority-class examples can remain hard under a classifier dominated by head-class gradients, but the focal term does not directly encode the class counts. It is therefore a difficulty-sensitive loss rather than a sampler or explicit prior correction.

3.5 Hybrid Sampling–Loss Methods

Hybrid methods combine data-level and algorithm-level changes. The Scholz-style representatives are placed here because the tested variants use class-balanced oversampling together with metric-derived losses. This coupling matters: their effect cannot be attributed only to a new loss or only to a new sampler.

3.5.1 Scholz-Style CE + Soft F1

Scholz et al. motivate losses derived from metrics that are already used to evaluate imbalanced medical classification, including F1 and Matthews correlation coefficient (MCC) (Scholz et al., 2024). The intuition is that CE optimizes the log-probability of the correct class one example at a time, whereas macro F1 evaluates a different object: the balance between precision and recall at class level. In a long-tailed dataset, this distinction matters because a model can reduce CE substantially by fitting frequent classes while still leaving tail-class recall or precision weak.

The obstacle is that ordinary F1 is computed from hard counts after prediction, so it is not directly usable as a gradient-based training loss. The Scholz-style soft-F1 objective replaces those hard counts with probability-weighted counts inside each minibatch. For one class, a confident correct prediction contributes almost one soft true positive, a confident prediction for the wrong class contributes to soft false positives, and low probability on a true class contributes to soft false negatives. Formally, for one-hot labels y_i and predicted probabilities $\hat{y}_i$:

$$\mathrm{TP} = \sum_i y_i \hat{y}_i, \quad \mathrm{FP} = \sum_i (1 - y_i)\hat{y}_i, \quad \mathrm{FN} = \sum_i y_i (1 - \hat{y}_i). \tag{7}$$

These terms keep the familiar meaning of a confusion matrix but make every component differentiable with respect to the model probabilities. They then define soft precision, soft recall, and soft F1:

$$\mathrm{F1}_{\mathrm{soft}} = \frac{2\,\mathrm{precision}_{\mathrm{soft}}\,\mathrm{recall}_{\mathrm{soft}}}{\mathrm{precision}_{\mathrm{soft}} + \mathrm{recall}_{\mathrm{soft}}}. \tag{8}$$

The multiclass implementation applies this construction separately to each class and averages the resulting classwise terms. This macro-style averaging is the imbalance-aware part: each class contributes one term regardless of how many samples it has in the full dataset. The tested objective does not replace CE completely; it adds the metric-derived term to CE so that optimization still rewards calibrated class probabilities while also pushing the minibatch toward higher macro soft F1:

$$\mathcal{L}_{\mathrm{CE+F1}} = \mathcal{L}_{\mathrm{CE}} + \left(1 - \frac{1}{C}\sum_{c=1}^{C} \mathrm{F1}_{\mathrm{soft},c}\right). \tag{9}$$

The benchmark trains this loss with balanced sampling on the patch manifest. That design is important for interpretation: the loss tries to optimize a macro-style class metric, while the sampler ensures that tail classes occur often enough in minibatches for their soft-F1 terms to provide a regular training signal.

3.5.2 Scholz-Style CE + Soft MCC

The soft-MCC variant uses the same principle but starts from a different evaluation metric. MCC is often used in imbalanced classification because it summarizes all cells of the confusion matrix rather than emphasizing only the positive class or only accuracy. In multiclass form, it can be read as a normalized agreement between the predicted class distribution and the true class distribution. A high value requires more than frequent-class accuracy: the predicted mass must line up with the correct classes without collapsing onto the head classes.

To make MCC trainable, the hard predicted labels are again replaced by probabilities. Let $Y \in \{0,1\}^{N\times C}$ be the one-hot label matrix for a minibatch and $\widehat{Y} \in [0,1]^{N\times C}$ the predicted probability matrix. The numerator measures probability-weighted agreement after subtracting agreement expected from the marginal class totals. The denominator normalizes by the spread of the predicted and true class totals, so the quantity remains comparable across minibatches:

$$\mathrm{MCC}_{\mathrm{soft}} = \frac{N\sum_{i,c} Y_{ic}\widehat{Y}_{ic} - \sum_c \left(\sum_i Y_{ic}\right)\left(\sum_i \widehat{Y}_{ic}\right)}{\sqrt{N^2 - \sum_c \left(\sum_i \widehat{Y}_{ic}\right)^2}\sqrt{N^2 - \sum_c \left(\sum_i Y_{ic}\right)^2}}. \tag{10}$$

As with soft F1, the training objective turns a score to be maximized into a loss to be minimized and adds it to CE:

$$\mathcal{L}_{\mathrm{CE+MCC}} = \mathcal{L}_{\mathrm{CE}} + (1 - \mathrm{MCC}_{\mathrm{soft}})\,. \tag{11}$$

This variant is also evaluated only in the patch benchmark and is trained with class-balanced oversampling. The soft-F1 and soft-MCC variants therefore ask a focused question: once tail classes are regularly presented to the optimizer, does adding a differentiable version of a class-balanced evaluation metric improve the controlled patch classifier beyond CE and simple balanced sampling?

3.6 Synthetic Data Generation

Synthetic generation is data-level in spirit but is separated because it introduces new generated samples rather than resampling observed ones. In histopathology, this family is attractive because deleting majority-class tissue wastes data, while naive image-space interpolation often produces unrealistic pathology images (Saini and Susan, 2023; Ruiz-Casado et al., 2024). The practical hope is simple: if real minority-class patches are scarce, a generator may learn enough of their stain, texture, and tissue morphology distribution to provide additional plausible training examples.

3.6.1 ProGAN Augmentation

The ProGAN experiment adapts the GAN-based histopathology augmentation idea of Ruiz-Casado et al. to multiclass TCGA-UT patches (Ruiz-Casado et al., 2024). A generative adversarial network (GAN) has two coupled models. The generator G maps random latent vectors to synthetic images, while the discriminator D tries to distinguish real training patches from generated patches. Training alternates between making D better at this distinction and making G produce images that D cannot reliably reject. The standard objective expresses this competition as

$$\min_G \max_D \mathbb{E}_{\mathbf{x}\sim p_{\mathrm{data}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{u}\sim p(\mathbf{u})}[\log(1 - D(G(\mathbf{u})))]. \tag{12}$$

This equation is not a classification loss for the final TCGA-UT classifier. It is the pretraining objective for producing additional patch images. After generation, the classifier simply sees a larger patch training set.

ProGAN modifies ordinary GAN training by growing the image resolution gradually. Instead of asking the generator to synthesize 256×256 histology patches from the start, training begins at a coarse resolution where global color and texture structure is easier to model. Higher-resolution layers are then introduced step by step until the target patch size is reached. This staged procedure is intended to stabilize synthesis and reduce the risk that the generator learns only a narrow set of repeated images. In this benchmark, one class-specific generator is trained for every training class whose patch count is below the head-class training count. Progressive training uses seven resolution stages from 4×4 to 256×256 pixels, with 25 epochs per stage, a fade-in fraction of 0.5 for newly added layers, latent dimension 128, base channel width 512, Adam optimization with learning rate 2×10^{-4} and $\beta_1 = 0.5$, and the depth-dependent batch schedule of Ruiz-Casado et al. When a class has more than 2048 real training patches, at most 2048 are drawn uniformly at random with a deterministic per-class seed for ProGAN training only; the downstream classifier still uses the full real training set together with the generated patches. Synthetic patch counts are set to the head-class training count minus the class training count, so each augmented tail class matches the largest real training class. Inception FID is computed when `torchvision` is available and stored as a diagnostic, but no generated samples are filtered by FID or other quality thresholds. Real and synthetic RGB patches are embedded with the same frozen Virchow2 encoder before classifier training, so the augmentation is evaluated in the same embedding regime as the other patch methods. Synthetic images are used only as extra patch-classification training examples; they are not converted into synthetic WSI bags.

3.7 Representation and Architecture-Level Methods

Representation-level methods try to make rare classes more separable in feature space rather than only changing sampling or per-example loss weights. This family is especially relevant for computational pathology, where WSI labels supervise bags of many unlabeled instances and class imbalance can occur both across slides and within a positive slide (Juyal et al., 2024).

3.7.1 SC-MIL

SC-MIL extends supervised contrastive learning to MIL by applying the contrastive objective to bag representations instead of assigning noisy slide labels to individual patches (Juyal et al., 2024). The motivation is that a slide label usually does not describe every patch in a bag. A cancer-type label can be correct for the slide while many instances contain background, normal tissue, artifacts, or weakly informative morphology. Applying contrastive learning directly to instances would therefore treat many patches as positives or negatives using labels they do not truly carry. SC-MIL avoids that mismatch by contrasting the aggregated slide-level representation.

The attention-MIL backbone first maps bag B_i to an embedding $\mathbf{b}_i = \mathbf{b}(B_i)$. A projection head g then produces normalized contrastive features $\mathbf{r}_i = g(\mathbf{b}_i)$. In the contrastive view, each bag in a minibatch is an anchor. Bags with the same cancer-type label are positives because their slide-level representations should become closer, while bags from other classes act as negatives because their representations should remain separated. For anchor i, bags of the same class form the positive set $P(i)$, and the supervised contrastive term is

$$\mathcal{L}_{\mathrm{SCL}} = -\frac{1}{\sum_i |P(i)|} \sum_i \sum_{p \in P(i)} \log \frac{\exp(\mathbf{r}_i^\top \mathbf{r}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{r}_i^\top \mathbf{r}_a / \tau)}. \tag{13}$$

The numerator rewards similarity to same-class bags. The denominator compares that similarity against all other bags in the minibatch, so the loss favors representations where class clusters are compact and mutually separated. The temperature τ controls how sharply these similarities are weighted; this benchmark uses $\tau = 1.0$.

SC-MIL does not rely on contrastive learning alone. The contrastive term shapes the representation space, while CE trains the final class predictor. The tested variant combines them through a linear curriculum:

$$\mathcal{L}_t = \beta_t \mathcal{L}_{\mathrm{SCL}} + (1 - \beta_t)\mathcal{L}_{\mathrm{CE}}, \qquad \beta_t = 1 - \mathrm{progress}_t. \tag{14}$$

Early training therefore emphasizes class-structured bag embeddings, when the classifier head is still unstable. As training progresses, β_t decreases and more weight moves to CE, so the learned representation is tied back to the actual slide-level classification objective. Activation diagnostics count positive pairs during training to confirm that the contrastive component is active.

3.8 Training Protocol and Reproducibility

All benchmark hyperparameters are fixed in the experiment configuration and are not tuned on the validation split. Validation metrics are written for inspection, but model selection, early stopping, and test reporting all use the weights after the final training epoch. Each benchmark is repeated over seeds 0, 1, and 2, which fix both the slide partition and the PyTorch random seeds used for initialization, shuffling, and sampling.

Table 3 summarizes the shared optimization settings. Patch and WSI-bag classifiers use AdamW with learning rate 10^{-3}, weight decay 10^{-4}, and 30 fixed epochs without early stopping. Patch training uses minibatches of 256 embeddings; WSI-bag training uses minibatches of 32 slide bags. Focal loss uses $\gamma = 2.0$, RankMix uses $\alpha = 1.0$, and SC-MIL uses contrastive temperature $\tau = 1.0$. These method-specific constants follow the cited references or common defaults rather than a validation sweep.

Table 3: Shared training protocol for both benchmarks. All methods within a benchmark use the same optimizer, schedule, and epoch budget unless noted.

Setting	Patch benchmark	WSI-bag benchmark
Optimizer	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}
Epochs	30	30
Early stopping	none	none
Minibatch size	256 patches	32 slide bags
Model selection	final epoch	final epoch
Random seeds	0, 1, 2	0, 1, 2

The patch classifier head is a two-layer MLP on frozen 2560-dimensional Virchow2 embeddings: a linear map to 512 units, ReLU, dropout with rate 0.1, and a linear output layer to 32 classes. The WSI-bag model is a gated attention MIL network on the same 2560-dimensional instance features. Each instance is encoded with a linear map to 256 units followed by ReLU and dropout 0.1; instance attention weights come from a linear scorer followed by a softmax over instances; the bag representation is the attention-weighted sum of instance embeddings; and slide logits are produced by a linear map from the 256-dimensional bag embedding to 32 classes. SC-MIL adds a projection head with two linear layers of width 256 and ReLU between them. No cap is applied to the number of instances per WSI bag in the reported benchmark.

RankMix uses balanced slide-bag sampling during student training. Mixed bags are formed within each minibatch by pairing bags with a random permutation and mixing their teacher-ranked representative subsequences with $\lambda \sim \mathrm{Beta}(1, 1)$. The student is reinitialized after teacher preparation and trained for the same 30-epoch budget as the other MIL methods.

Head, body, and tail summaries group classes by support in the full TCGA-UT slide distribution rather than by held-out test counts. Cancer types are sorted by dataset slide count; the eight lowest-support classes form the tail tier, the eight highest-support classes form the head tier, and the remaining sixteen classes form the body tier. Tier-wise precision, recall, and F1 on validation and test splits then aggregate only over the predefined classes in each tier. Expected calibration error uses ten equal-width confidence bins on the maximum predicted probability.

4 Evaluation

Both benchmarks use seeds 0, 1, and 2. Macro F1 is the primary ranking metric within each benchmark because it weights classes equally and penalizes failure on rare classes. Balanced accuracy, accuracy, classwise precision/recall/F1, and head/body/tail summaries are reported in the main result tables using the tier definitions in Section 3.8. Negative log likelihood (NLL), multiclass Brier score, and expected calibration error (ECE) are computed on held-out softmax probabilities and summarized in Table 8. Patch and WSI-bag methods are never ranked against each other, because they answer related but different questions with different input units and supervision.

All reported tables use the held-out test split and show mean $\pm$ standard deviation over seeds. Table 7 additionally reports paired seed-wise differences for the headline comparisons against each regime's CE baseline; bracketed values list deltas in seed order 0–2 without formal significance testing. Separate validation summaries are retained as artifacts for model-selection checks. The completed sweep contains no missing planned results for either benchmark.

5 Results

Both benchmarks share slide-disjoint splits and the Virchow2 feature family, but they answer different prediction questions under different per-slide evidence budgets. The patch benchmark treats each selected crop as an independent labeled example and caps slides at 30 deterministic patches; the WSI-bag benchmark predicts one slide label from an uncapped attention-weighted aggregation over all feature instances on that slide. Patch metrics therefore characterize patch-level classification only and must not be read as slide-level diagnostics; WSI-bag metrics characterize slide-level MIL only. Absolute scores and method orderings are not comparable across the two tables.

5.1 Patch-Level Results

Table 4: Patch-level test results over three seeds. Methods share the same controlled manifest, frozen Virchow2 encoder, linear classifier head, and evaluation code.

Method	Accuracy	Balanced accuracy	Macro F1
ProGAN augmentation	0.540 ± 0.002	0.441 ± 0.003	0.446 ± 0.003
Focal	0.419 ± 0.006	0.318 ± 0.004	0.323 ± 0.001
CE	0.424 ± 0.022	0.313 ± 0.021	0.319 ± 0.019
Balanced sampler	0.365 ± 0.010	0.356 ± 0.011	0.307 ± 0.012
CE + soft F1 (balanced)	0.356 ± 0.022	0.349 ± 0.023	0.298 ± 0.018
CE + soft MCC (balanced)	0.352 ± 0.026	0.347 ± 0.008	0.296 ± 0.009
Weighted CE	0.347 ± 0.013	0.340 ± 0.013	0.289 ± 0.014

The patch benchmark compresses into a narrow performance band once patches are represented with frozen Virchow2 embeddings (Table 4). The Scholz-style CE plus soft-MCC objective with balanced sampling ranks highest by macro F1 (0.782 ± 0.007) and balanced accuracy (0.782 ± 0.008). The closely related CE plus soft-F1 variant reaches nearly the same macro F1 (0.782 ± 0.004) and balanced accuracy (0.782 ± 0.004). Relative to plain CE (0.776 ± 0.002 macro F1, 0.770 ± 0.003 balanced accuracy), the hybrid objectives provide small but consistent improvements over CE in the frozen-embedding patch regime (Table 7): paired seed-wise macro-F1 deltas are positive for both Scholz-style variants in all three seeds, with mean gains of about +0.006–+0.007 points. The same methods do not improve probability quality on the same scale (Table 8): CE plus soft F1 raises NLL to 2.20 ± 0.09 and CE plus soft MCC to 1.95 ± 0.09, compared with 1.81 ± 0.09 for plain CE, while expected calibration error remains near 0.14 for both hybrid objectives and the baseline.

The simpler patch interventions separate more clearly in this embedding regime. Class-weighted CE and balanced sampling remain competitive with CE, with macro F1 values of 0.776 ± 0.012 and 0.775 ± 0.008, respectively. ProGAN augmentation does not outperform the CE baseline: it reaches macro F1 0.774 ± 0.008 and balanced accuracy 0.767 ± 0.011, which is within the spread of the non-generative methods rather than above them. This negative result is not explained by a lack of synthetic volume alone. For seed 0, ProGAN adds 348,120 generated training patches across 31 classes below the head-class patch count (Table 5) on top of the same real controlled training set as plain CE. Quality diagnostics show high Inception FID (median 178.5) and non-trivial Virchow2 nearest-neighbor distances (median 47.9 in L_2 on 2560-dimensional embeddings), indicating that the generated patches are neither near-exact copies of real training patches nor close to the real feature manifold used at classification time (Figure 4). Focal loss is the weakest patch method (0.766±0.003 macro F1, 0.760±0.002 balanced accuracy), indicating that down-weighting easy examples is not helpful once the representation already separates most cancer types well. Accuracy remains high for all methods (≈ 0.83), so the ranking is driven mainly by class-balanced metrics rather than by overall prevalence-sensitive performance.

Table 5: ProGAN augmentation diagnostics for seed 0. "Generated" counts are synthetic patches added per class to reach the head-class training patch count. Inception FID compares generated patches to the real training subset used for GAN training. Virchow2 mean NN is the mean, over a fixed synthetic subsample, of the minimum L_2 distance to real same-class embeddings in the frozen feature cache.

Class	Real train	Generated	Inception FID	Virchow2 mean NN
Lymphoid Neoplasm Diffuse Large B-cell Lymphoma	600	16000	315.9	52.16
Cholangiocarcinoma	660	15940	232.2	49.09
Pheochromocytoma and Paraganglioma	930	15670	301.3	52.59
Uveal Melanoma	1160	15440	255.9	52.46
Rectum adenocarcinoma	1350	15250	178.5	51.31
Uterine Carcinosarcoma	1460	15140	161.8	49.41
Mesothelioma	1500	15100	249.1	46.97
Kidney Chromophobe	1740	14860	207.7	40.87
All augmented classes (31 total)	166480	348120	178.5 median	47.89 median

The classwise-recall plot (Figure 6) shows that remaining errors are distributed across many cancer types rather than concentrated in a single tail class. In this setting, the main practical distinction is therefore not whether imbalance mitigation works at all, but which mechanism best preserves rare-class recall when the feature extractor is already strong. These conclusions remain confined to the patch benchmark: high patch accuracy or macro F1 does not directly establish slide-level utility, because each test prediction is made on a single crop rather than by aggregating evidence across a slide bag.

5.2 WSI-Bag Results

Table 6: WSI-bag test results over three seeds. Methods share the same full feature bags, attention-MIL backbone family, and evaluation code.

Method	Accuracy	Balanced accuracy	Macro F1
RankMix	0.894 ± 0.010	0.863 ± 0.008	0.848 ± 0.017
MIL CE	0.890 ± 0.006	0.836 ± 0.020	0.840 ± 0.020
Balanced MIL	0.885 ± 0.007	0.837 ± 0.015	0.836 ± 0.013
SC-MIL	0.881 ± 0.007	0.837 ± 0.017	0.833 ± 0.016
Weighted MIL	0.880 ± 0.005	0.834 ± 0.018	0.829 ± 0.018
Focal MIL	0.879 ± 0.004	0.824 ± 0.011	0.827 ± 0.009

The WSI-bag benchmark evaluates one slide-level prediction per diagnostic slide by aggregating uncapped feature bags from the shared split. Although the median bag size in the frozen feature export is also about 30 instances, larger slides contribute up to 70 instances to MIL whereas the patch benchmark never uses more than 30 deterministic crops from the same slide, and MIL additionally learns instance weighting rather than treating each crop as

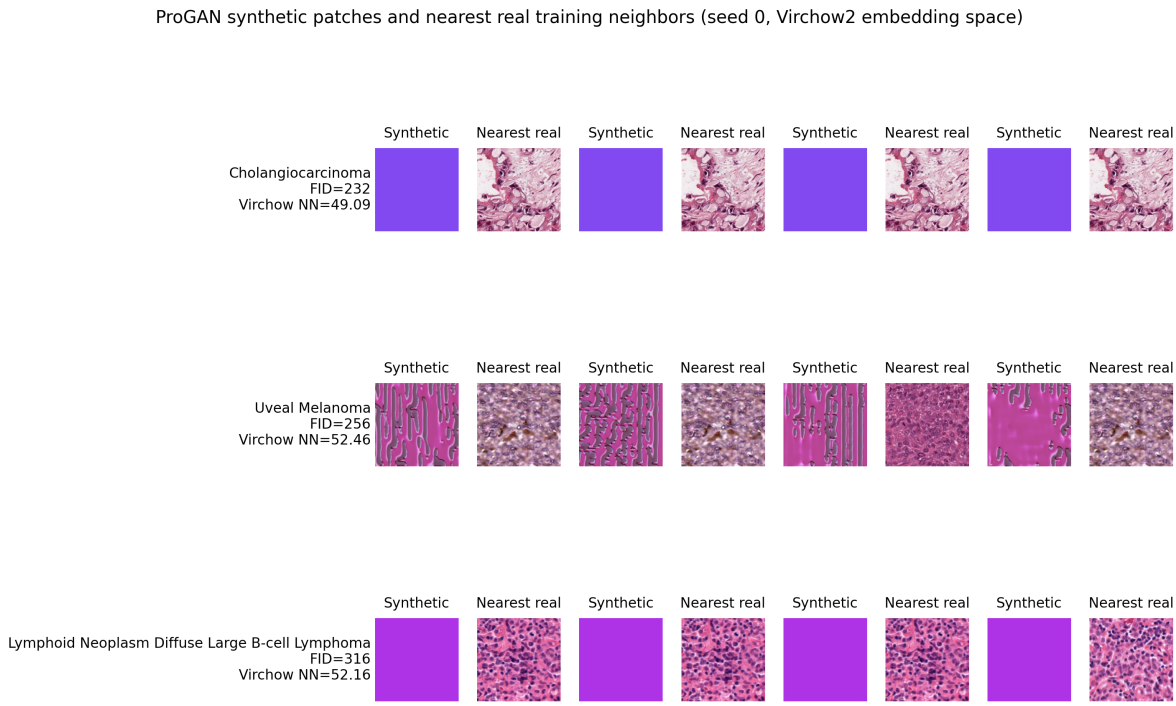


Figure 4: Example ProGAN synthetic patches (left column of each pair) and their nearest real training neighbors in Virchow2 embedding space (right column) for three tail classes in seed 0. The visual and embedding-space gaps motivate reporting FID and Virchow2 nearest-neighbor distance alongside classifier performance.

an independently labeled patch. WSI-bag performance may therefore benefit from both extra morphological evidence on large slides and slide-native aggregation mechanisms that the patch classifier does not exercise. Table 6 should be read as a within-regime slide-level leaderboard, not as confirmation or refutation of the patch-level ranking.

RankMix shows the clearest WSI-bag benefit, especially in balanced accuracy, while macro-F1 gains remain modest (Table 6). Relative to plain MIL CE, balanced accuracy improves from 0.836 ± 0.020 to 0.863 ± 0.008, whereas macro F1 moves from 0.840 ± 0.020 to 0.848 ± 0.017 with overlapping standard deviations across seeds. Paired seed-wise comparisons (Table 7) show positive RankMix deltas in all three seeds for both metrics, with the largest balanced-accuracy shift in seed 0 (+0.045). RankMix also improves probability calibration (Table 8), lowering NLL from 1.03 ± 0.05 to 0.41 ± 0.04, Brier score from 0.195 ± 0.009 to 0.156 ± 0.015, and ECE from 0.089 ± 0.003 to 0.022 ± 0.003. The larger balanced-accuracy shift is consistent with RankMix directly augmenting representative instance subsequences in the MIL regime.

The remaining WSI methods cluster closely. Balanced-sampling MIL and SC-MIL produce similar balanced accuracy (0.837), but both trail RankMix in macro F1. SC-MIL performs well on validation but does not exceed plain MIL CE on the test macro-F1 average. Weighted MIL CE and focal MIL do not improve over plain MIL CE in this configuration, with focal MIL giving the lowest WSI macro F1. Activation diagnostics confirm that the advanced WSI objectives were active: RankMix generated about 1.10 million mixed examples per seed, and SC-MIL constructed about 1.06 million positive pairs per seed.

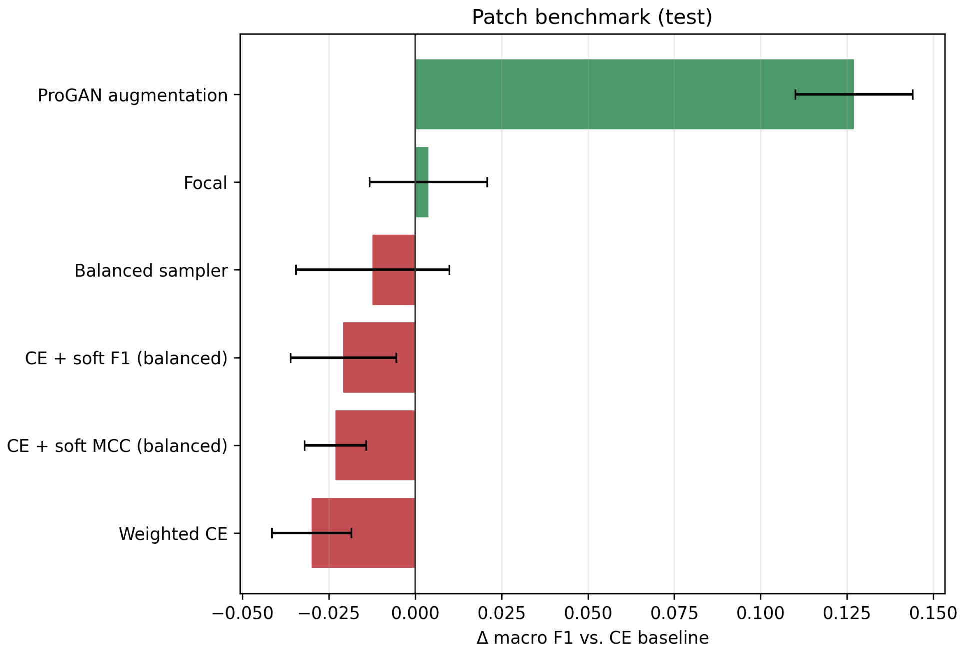


Figure 5: Patch-level change in macro F1 relative to the CE baseline on the held-out test split. Bars show the mean Δ over seeds 0–2 with seed-wise standard deviation; values should only be interpreted within the patch benchmark (Table 4).

5.3 Paired Seed Comparisons

Table 7: Paired test-split differences against the within-regime CE baseline on seeds 0–2. Each cell reports mean ± SD across seeds; bracketed values list seed-ordered paired deltas. No formal p-values are computed.

Comparison	Δ Macro F1	Δ Balanced accuracy
CE + soft MCC − CE	-0.023 ± 0.009 [−0.027, −0.011, −0.031]	$+0.034 \pm 0.012$ [+0.024, +0.051, +0.029]
CE + soft F1 − CE	-0.021 ± 0.015 [−0.042, −0.005, −0.015]	$+0.037 \pm 0.023$ [+0.005, +0.057, +0.048]
RankMix − MIL CE	$+0.008 \pm 0.006$ [+0.017, +0.005, +0.003]	$+0.027 \pm 0.013$ [+0.045, +0.015, +0.020]

Table 7 makes the headline gains explicit at seed resolution rather than only through aggregate means. Both patch hybrid objectives remain above plain CE on macro F1 in every seed, and RankMix remains above plain MIL CE on both metrics in every seed, even though the absolute shifts are modest and would require formal paired testing to treat as significant.

5.4 Probability Calibration

Table 8 makes the discrimination–calibration tradeoff explicit. On patches, focal loss yields the lowest ECE (0.013 ± 0.003) but the weakest macro F1, whereas the Scholz-style hybrid objectives that rank highest on macro F1 increase NLL without a matching ECE gain over plain CE. ProGAN augmentation further degrades both NLL and ECE relative to the CE baseline. On WSI bags, RankMix is the only method that improves both class-balanced discrimination and calibration relative to plain MIL CE; the remaining MIL variants cluster near the baseline on Brier score and ECE while trailing RankMix on both metrics.

All WSI-bag calibration scores are computed from the same held-out test slides, hard slide labels, and raw softmax outputs of the final-epoch model. Every method uses the same eval-

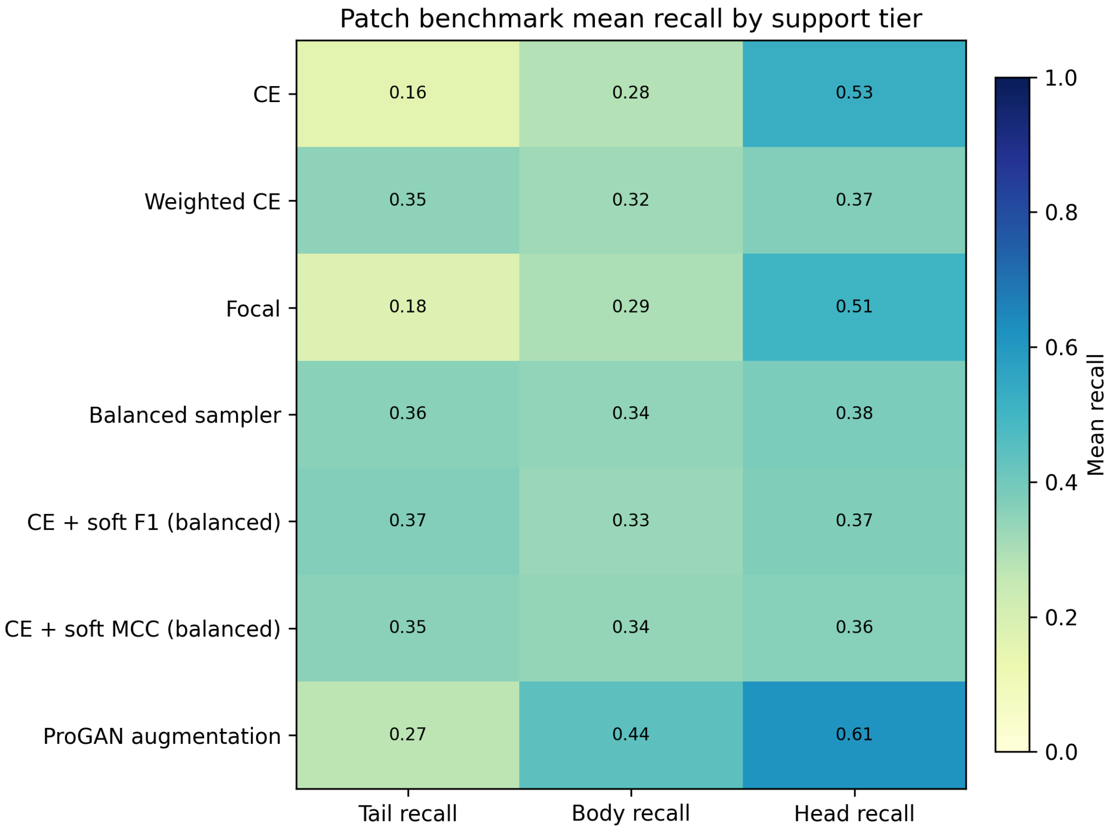


Figure 6: Patch-level classwise recall on the held-out test split. The plot exposes the class-level variation hidden by aggregate accuracy and motivates reporting balanced accuracy and macro F1.

uation routine: no temperature scaling, no validation-fit calibration, and no method-specific post-processing. An audit over stored test-result files confirms that all six WSI methods share identical test label sequences within each seed, that reported probabilities sum to one, and that stored NLL, Brier, and ECE values match recomputation from the saved probability matrix. The RankMix gap is therefore not an evaluation artifact: plain MIL CE is strongly overconfident (mean maximum predicted probability ≈ 0.98 versus ≈ 0.91 for RankMix), whereas RankMix's soft mixing targets affect training only and not the held-out evaluation forward pass.

Table 8: Held-out test calibration metrics over three seeds. Lower is better for negative log likelihood (NLL), multiclass Brier score, and expected calibration error (ECE). Values are not comparable across regimes because patch and WSI-bag models operate on different input units.

Regime	Method	NLL	Brier	ECE
Patch	ProGAN augmentation	1.802 ± 0.023	0.610 ± 0.008	0.092 ± 0.015
Patch	Focal	2.122 ± 0.018	0.725 ± 0.007	0.013 ± 0.003
Patch	CE	2.101 ± 0.056	0.719 ± 0.021	0.034 ± 0.004
Patch	Balanced sampler	2.401 ± 0.017	0.785 ± 0.011	0.094 ± 0.014
Patch	CE + soft F1 (balanced)	2.434 ± 0.087	0.791 ± 0.024	0.094 ± 0.011
Patch	CE + soft MCC (balanced)	2.502 ± 0.083	0.805 ± 0.027	0.129 ± 0.001
Patch	Weighted CE	2.373 ± 0.033	0.790 ± 0.014	0.072 ± 0.015
WSI bag	RankMix	0.407 ± 0.041	0.156 ± 0.015	0.022 ± 0.003
WSI bag	MIL CE	1.034 ± 0.051	0.195 ± 0.009	0.089 ± 0.003
WSI bag	Balanced MIL	1.100 ± 0.050	0.201 ± 0.009	0.091 ± 0.003
WSI bag	SC-MIL	0.958 ± 0.104	0.205 ± 0.013	0.092 ± 0.005
WSI bag	Weighted MIL	1.124 ± 0.028	0.210 ± 0.011	0.096 ± 0.005
WSI bag	Focal MIL	0.763 ± 0.064	0.200 ± 0.007	0.078 ± 0.006

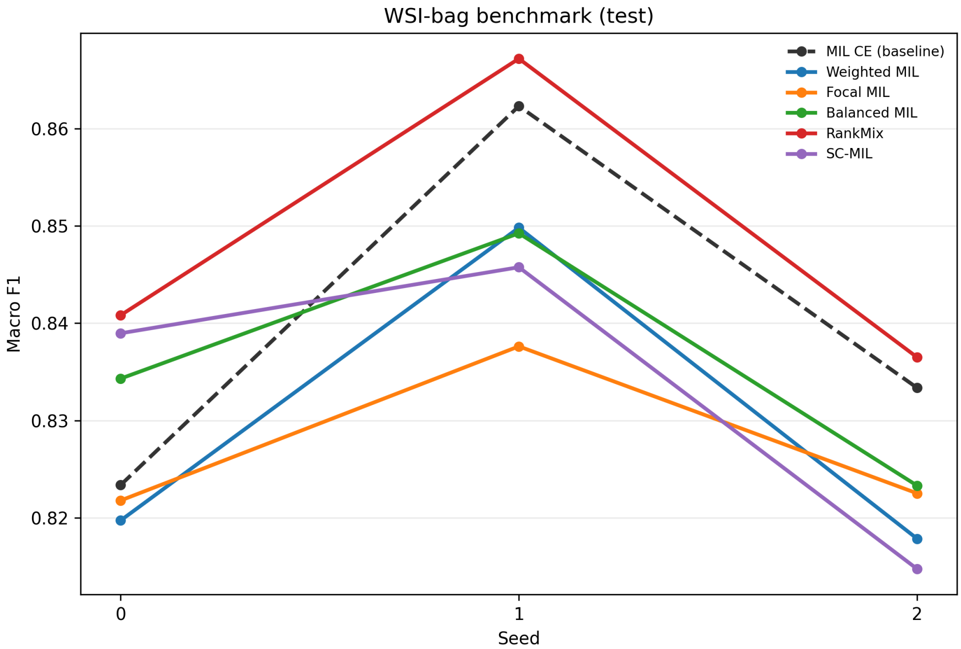


Figure 7: WSI-bag macro F1 on the held-out test split by random seed. The dashed line marks plain MIL CE; seed-wise spread shows why cross-method gaps should be read cautiously when standard deviations overlap (Table 6).

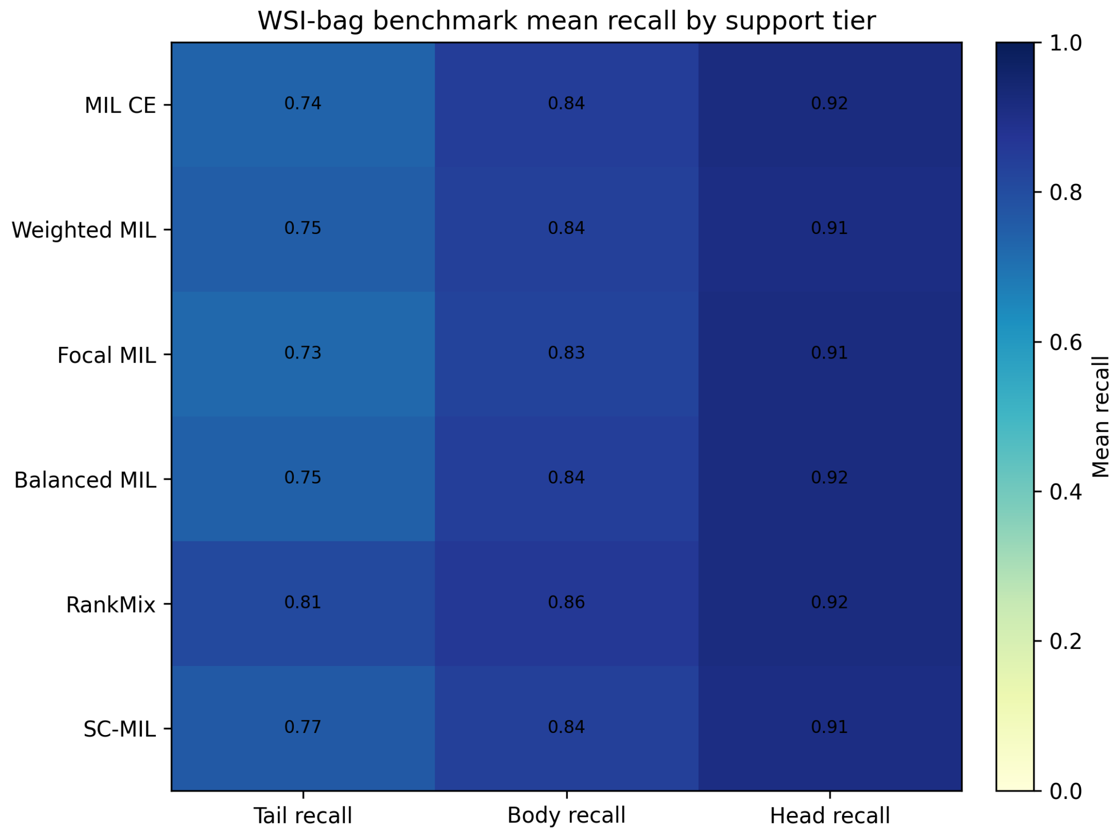


Figure 8: WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient for the long-tailed setting.

6 Discussion

The two benchmarks support different but compatible conclusions. In the patch benchmark, a strong frozen encoder compresses the performance spread: most methods cluster near macro F1 0.77–0.78, and the Scholz-style hybrid objectives provide small but consistent improvements over plain CE rather than large separations. ProGAN augmentation does not deliver the same advantage in this regime even though it adds large numbers of synthetic tail-class patches; the accompanying quality diagnostics suggest a combination of high Inception FID and substantial Virchow2 embedding-space separation from real training patches, rather than a simple failure to increase tail-class exposure. Focal loss is also unhelpful here, consistent with the interpretation that difficulty reweighting is redundant when the representation has already absorbed much of the class-separation problem. In the WSI-bag benchmark, RankMix shows the clearest benefit, especially in balanced accuracy, while macro-F1 gains remain modest; a WSI-native feature-mixing mechanism can still help over plain MIL CE even though the bags already use Virchow2 features.

These readings should be interpreted within the benchmark's deliberate scope. The study is a controlled comparison rather than a full reproduction of every original paper: hyperparameters follow fixed configuration values and cited defaults rather than method-specific validation tuning, and both regimes use frozen Virchow2 features rather than end-to-end encoder adaptation. Rankings therefore describe how imbalance mechanisms behave when patch and slide classification are already supported by a strong pathology foundation model, not every published implementation under its preferred training recipe. The ProGAN adaptation likewise keeps the progressive-generator mechanism from Ruiz-Casado-style histology augmentation but applies it to multiclass TCGA-UT rather than reproducing the full GAN-family comparison from the source work.

Holding the encoder fixed within each benchmark therefore changes what an imbalance intervention can accomplish. On patches, the bottleneck shifts from learning stain and tissue morphology to calibrating decision boundaries under a long-tailed label distribution. Under that bottleneck, objectives that approximate macro F1 or MCC on balanced minibatches are better matched to the evaluation target than image-space synthesis or focal down-weighting of easy examples, but they can do so at the cost of worse likelihood scores without clear ECE improvements over CE. On slides, the bottleneck instead includes instance selection and bag aggregation, so RankMix can still add value by mixing informative subsequences even when the underlying features are strong, and in this benchmark it improves calibration as well as balanced accuracy.

This pattern reinforces the need to evaluate imbalance methods in their native input regimes and at the stage where the model is actually limited. A patch generator should not be judged by first converting synthetic images into one-instance MIL bags, and a MIL feature-mixing method should not be compared directly against a patch-level linear head on single embeddings. The shared split and encoder family link the two benchmarks descriptively, but they do not license cross-regime inference: patch accuracy does not directly imply slide-level deployment quality, WSI-bag gains may partly reflect access to more per-slide evidence and aggregation behavior absent from the capped patch manifest, and neither absolute scores nor method orderings should be transferred across tables. The controlled design narrows the claims but makes them easier to defend: within each benchmark, the split, encoder family, head architecture, input budget, and evaluation code are shared.

Several limitations remain. TCGA-UT patch labels come from pathologist-selected tumor regions; they are useful patch-level class labels but not dense annotations. The two benchmarks share slide-disjoint splits but differ in prediction unit and per-slide evidence budget, so their result tables remain non-transferable as discussed above.

7 Conclusion

TCGA-UT supports a cleaner imbalance study when patch-level and WSI-bag supervision are evaluated separately with regime-appropriate inputs. On controlled patch classification with frozen Virchow2 embeddings, Scholz-style CE plus soft F1 or soft MCC with balanced sampling provide small but consistent improvements over plain CE, whereas ProGAN augmentation and focal loss do not improve over that baseline. On WSI bags, RankMix shows the clearest benefit, especially in balanced accuracy, while macro-F1 gains remain modest. These two result sets are complementary rather than mutually confirming: patch-level metrics do not directly establish slide-level utility, WSI-bag metrics may benefit from uncapped per-slide evidence and MIL aggregation beyond the 30-patch controlled manifest, and methods should not be ranked across regimes. The practical implication is that imbalance mitigation should be matched to the supervision unit and to the learning stage that still limits performance; otherwise, an intervention can appear ineffective because the representation already absorbs the problem it was designed to solve, or because it was tested in the wrong regime.

References

Chen, Y.-C. and Lu, C.-S. (2023). Rankmix: Data augmentation for weakly supervised learning of classifying whole slide images with diverse sizes and imbalanced categories. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23936–23945.

Juyal, D., Shingi, S., Javed, S. A., Padigela, H., Shah, C., Sampat, A., Khosla, A., Abel, J., and Taylor-Weiner, A. (2024). SC-MIL: Supervised contrastive multiple instance learning for imbalanced classification in pathology. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 7016–7025.

Komura, D., Kawabe, A., Fukuta, K., Sano, K., Umezaki, T., Koda, H., Suzuki, R., Tominaga, K., Ochi, M., Konishi, H., et al. (2022). Universal encoding of pan-cancer histology by deep texture representations. *Cell Reports*, 38(9):110424.

Mosquera, C., Ferrer, L., Milone, D. H., Luna, D., and Ferrante, E. (2024). Class imbalance on medical image classification: towards better evaluation practices for discrimination and calibration performance. *European Radiology*, 34(12):7895–7903.

Ruiz-Casado, J. L., Molina-Cabello, M. A., and Luque-Baena, M. R. (2024). Enhancing histopathological image classification performance through synthetic data generation with generative adversarial networks. *Sensors*, 24(12):3777.

Saini, M. and Susan, S. (2023). Tackling class imbalance in computer vision: a contemporary review. *Artificial Intelligence Review*, 56(S1):1279–1335.

Scholz, D., Erdur, A. C., Buchner, J., Peeken, J. C., Rueckert, D., and Wiestler, B. (2024). Imbalance-aware loss functions improve medical image classification. In *Proceedings of the 7th International Conference on Medical Imaging with Deep Learning*, volume 250 of *Proceedings of Machine Learning Research*, pages 1341–1356. PMLR.

Tomczak, K., Czerwinska, P., and Wiznerowicz, M. (2015). The cancer genome atlas (tcga): an immeasurable source of knowledge. *Contemporary Oncology*, 19(1A):A68–A77.

Vorontsov, E., Bozkurt, A., Casson, A., Shaikovski, G., Zelechowski, M., Severson, K., Zimmermann, E., et al. (2024). A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30(10):2924–2935.

Zhang, Y., Kang, B., Hooi, B., and Yan, S. (2023). Deep long-tailed learning: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(9):10795–10816.

Zimmermann, E., Vorontsov, E., Viret, J., Casson, A., Zelechowski, M., and Shaikovski, G. (2024). Virchow2. *2024 Shell TechXplorer Digest*, 2024(1).